MATHS MANTHAN
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Subject: Mathematics **WEEKLY CHAPTER TEST – 03 (QUADRATIC EQUATIONS)** **Time:** 45 Minutes

Class: Class X (CBSE) **Chapter 4: Quadratic Equations** **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 10(iii).
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 VSAQ of 2 marks and 3 SA questions of 3 marks each.
4. **Section C** comprises 1 Applied Case Study with 3 sub-parts of 4 marks total.

SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

Q1. If one root of the equation $4x^2 - 2x + (k-4) = 0$ is the reciprocal of the other, then $k =$: [1]
(A) -8 (B) 8 (C) 4 (D) -4

Q2. If the discriminant of the quadratic equation $3x^2 - 2x + c = 0$ is 16, then the value of c is: [1]
(A) 1 (B) -1 (C) 2 (D) -2

Q3. A quadratic equation whose one root is 2 and the sum of whose roots is zero, is: [1]
(A) $x^2 + 4 = 0$ (B) $x^2 - 4 = 0$ (C) $4x^2 - 1 = 0$ (D) $x^2 - 2 = 0$

Q4. If the equation $x^2 - ax + 1 = 0$ has two distinct real roots, then: [1]
(A) $|a| = 2$ (B) $|a| < 2$ (C) $|a| > 2$ (D) None of these

Q5. **Assertion (A):** If one root of $x^2 - 4x + 1 = 0$ is $2 + \sqrt{3}$, then the other root is $2 - \sqrt{3}$.
Reason (R): Irrational / surd roots of a quadratic equation with rational coefficients always occur in conjugate pairs. [1]
(A) Both A and R are true and R is correct explanation of A. (B) Both are true, but R is not correct explanation.
(C) A is true, R is false. (D) A is false, R is true.

SECTION B: Short Answer Type Questions (Part I)

Q6. Find the discriminant of the quadratic equation $2x^2 - 4x + 3 = 0$ and state the nature of its roots. [2]

Q7. Solve for x using the quadratic formula: [3]

$$9x^2 - 9(a+b)x + (2a^2 + 5ab + 2b^2) = 0$$

SECTION B: Short Answer Type Questions (Continued) (Total Sec B = 11 Marks)

Q8. A cottage industry produces a certain number of pottery articles in a day. It was observed that the cost of production of each article (in rupees) was 3 more than twice the number of articles produced on that day. If the total cost of production on that day was Rs. 90, find the number of articles produced and the cost of each article. [3]

Q9. The diagonal of a rectangular field is 60 m more than the shorter side. If the longer side is 30 m more than the shorter side, find the dimensions of the field. [3]

SECTION C: Applied Case Study Based Question (1 × 4 = 4 Marks)

Q10. Case Study: Graphic Design Studio & Banner Enlargement

A graphic designer is preparing an exhibition banner by enlarging a rectangular artwork photograph measuring 18 cm in length and 12 cm in width. The designer increases both length and width by a uniform margin of x cm on all sides. The enlarged banner has a surface area of 576 cm^2.

Based on the above geometric enlargement, answer the following questions:

(i) Express the new length and new breadth of the banner in terms of x. [1]

(ii) Formulate the quadratic equation in x in standard form representing the enlarged area. [1]

(iii) Find the value of x and the dimensions of the enlarged banner. [2]

OR

Calculate the percentage increase in area from the original photograph to the banner. [2]

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